What parts make up an atom?

An atom has three main parts: *protons, neutrons* and *electrons*.

Protons and neutrons make up the centre of an atom. The centre part of an atom is called the *nucleus*.

The electrons are outside the nucleus. They orbit around the nucleus at very fast speeds. Electrons are much smaller than protons or neutrons.

Protons and electrons have electrical charges.

- Each proton has a *positive* (+) charge
- Each electron has a *negative* (–) charge
- A neutron is *neutral*. It has no charge.

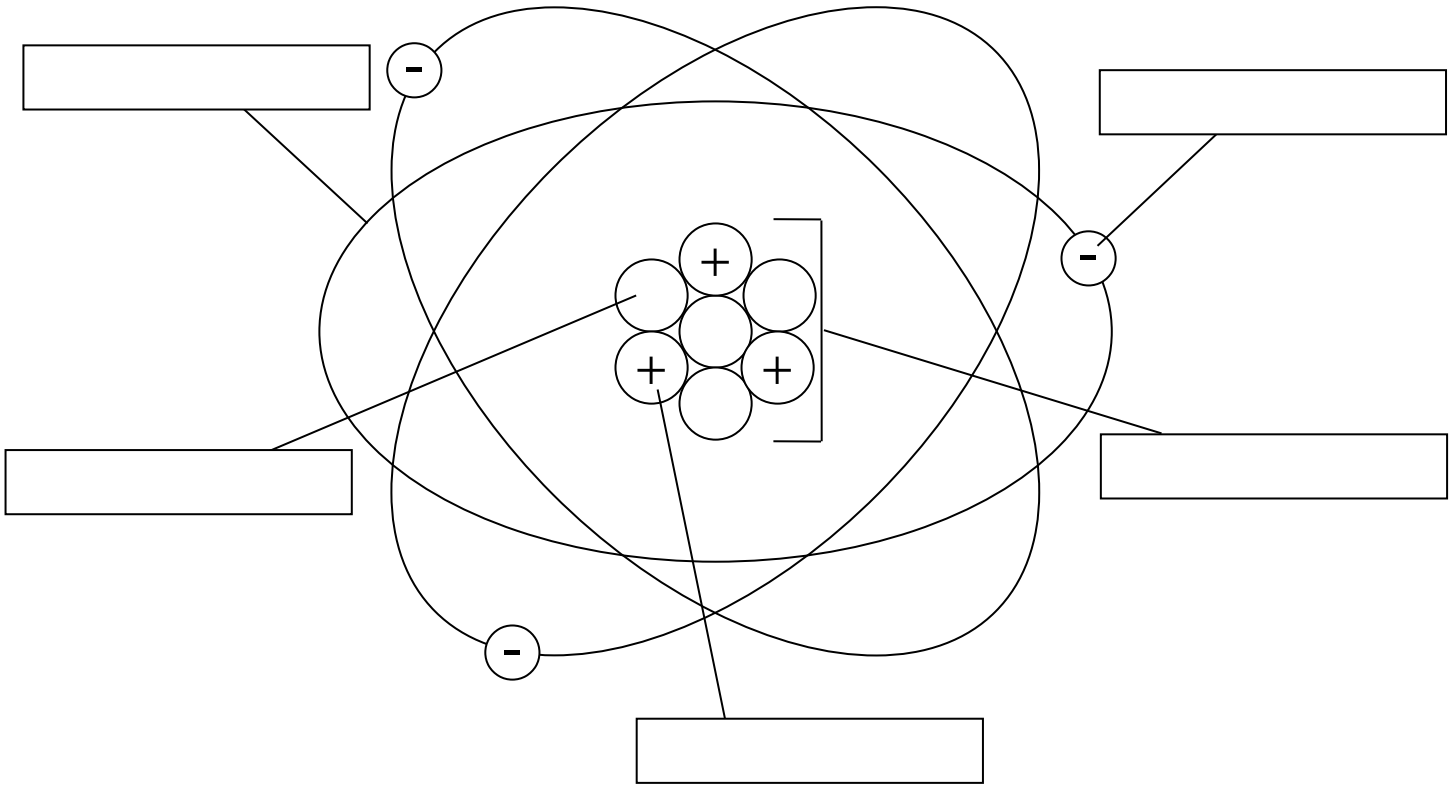
A neutral atom has the name number of protons as electrons.

PROTONS = ELECTRONS

This means that the number of positive charges equals the number of negative charges. They balance each other so the entire atom has no charge.

The *atomic number* is also the same as the number of protons.

Label the diagram of an atom with the following words: Electron, Orbit, Proton, Neutron, and Nucleus.



Fill in the missing words in the sentences below:

- The particles which make up atoms are called _____ , _____ and _____ .
- _____ and _____ are contained in the centre of the atom, which is called the _____ .
- _____ orbit around the _____.
- _____ are positively charged, _____ are negatively charged and _____ are neutral.

What is atomic mass?

Every atom has mass. The mass of an atom is called its *atomic mass*.

An atomic mass is not an exact scale reading of mass – like grams or kilograms. It is a way of comparing the mass of one atom against the mass of another atom.

The mass of an atom is the mass of its nucleus. A nucleus contains protons and neutrons, so the number of protons plus the number of neutrons gives us the mass of an atom.

Electrons are so light that they are not counted in the atomic mass.

Each proton is given a mass of one and each neutron is given a mass of one.

For example, a lithium atom has 3 protons and 4 neutrons. The atomic mass of a lithium atom is 7 (3 + 4 = 7)

The atomic mass is sometimes called the atomic weight.

Answer the following questions.

- What is the centre part of an atom called? _____
- What parts make up the centre of an atom?

_____

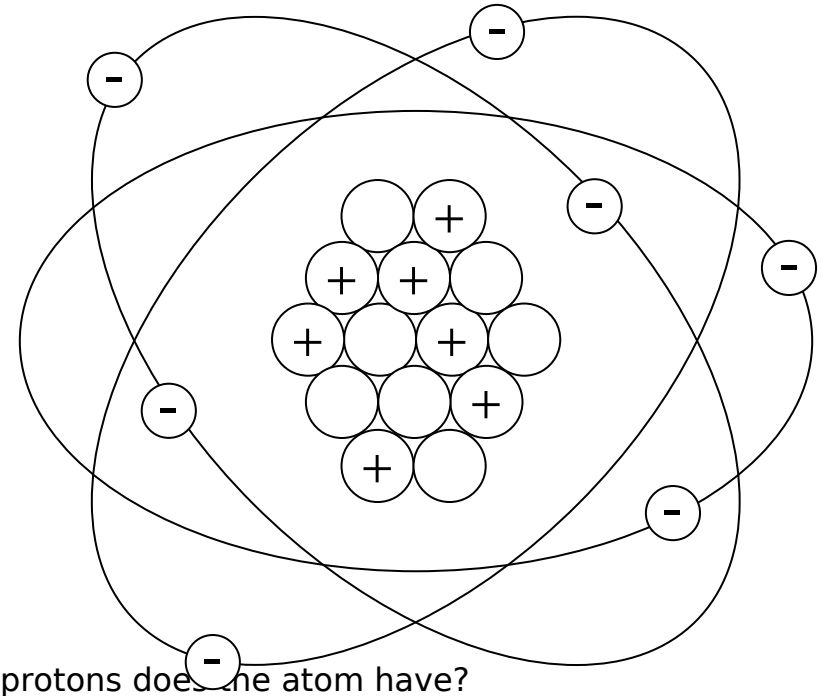
- The mass of an atom is the mass of its _____ and _____
- Fill in the blanks: ATOMIC MASS = _____ + _____
- An atom of Sodium has 11 protons and 12 neutrons. What is its atomic mass? _____

- An atom of fluorine has 9 protons and an atomic mass of 19. How many neutrons does it have? _____

Complete the following table without looking at a periodic table.

Name of element	Symbol	Atomic Number	Number of Protons	Number of Neutrons	Number of Electrons	Atomic Mass
Neon		10				20
Gold		79		118		
Chlorine				18		35
Nickel		28				59
Silicon		14				28
Sulphur					16	32
Oxygen				8	8	

What does the picture below show?



- How many protons does the atom have? _____
- How many neutrons does the atom have? _____
- How many electrons does the atom have? _____
- What is the atomic mass of this atom? _____
- What is the name of this atom? _____

- What is the symbol for this atom? _____